Open-Source Development

ITIC2262

Presented by:
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Year 2 Semester 2			
Course Code	ITIC2262		
Course Name	Open-Source Development		
Credit Value	2		
Core/Optional	Core		
Hourly Breakdown	Theory	Practical	Independent Learning
	15	30	55
Course Aim	To familiarize students with open-source software, methodologies, and architectures. Additionally, to provide hands-on expertise with open source products.		
Intended Learning Outcomes	After the completion of this module, the student will be able to, **LO1** Explain and elaborate open-source software development **LO2** Describe open-source software development concepts. **LO3** Working with the Open-source products. **LO4** Analyze features of the open-source products. **LO5** Examine the use of open-source product to improve business.		
Course Contents	**Introduction to source software development** introduction, Benefits, challenges and products.		**LO1**
	Describe open-source software development concepts software development process.		**LO2**
	Working with the Open-source products Identify opensource products specially OS, Office package, web etc.		**LO3** **LO4**
	Analyze features of the open-source products identify and use of the features of the open-source development.		**LO3** **LO4**
	Examine the use of open-source product to improve business		**LO4** **LO5**
Teaching/ Learning Methods	Lectures, demonstrations, practical sessions, learner-centred learning approaches, discussion, and group interactions.		
Assessment Strategy	Continuous and Semester end assessment		

Continuous Assessments 30%	**Assignments/ Tutorials (25%)**	• Assignment 1 – Open-source software development (5%) • Assignment 2 – Open-source Software development concepts (5%) • Assignment 3 – Working with open-source products (5%) • Assignment 4- Feature analysis (5%) • Assignment 5 – Use of Open-source products in different applications (5%)	**LO1-LO5**
	Mid-Term (5%)	• Mid semester examination 1 hour structured essay MCQ (5%)	**LO1-LO3**
Final Assessment 70 %		• Structured Essay 2h	**LO1-LO5**
References		1. M. N. RAO, FUNDAMENTALS OF OPEN SOURCE SOFTWARE, PHI Learning; 1st edition (September 16, 2014) 2. By Allen Tucker, Ralph Morelli, Chamindra de Silva . Software Development An Open Source Approach. CRC Press, 2011 ISBN 9781138051744	

Open Source Development

Build • Learn • Innovate

Better Code A Brighter Future

What is Open Source Development?

Open source development is the practice of creating software that is freely available, allowing anyone to use, study, modify and share the source code.

Linux Ubuntu GitHub VS Code Python



Collaborative
Work together across the globe

Flexible
Modify and customize to your needs

Cost Effective
No licensing fees

Innovative
Continuous improvement with community support

Learn
Explore source code and documentation

→

Code
Build and contribute to projects

→

Collaborate
Share knowledge and ideas

→

Improve
Create solutions for real-world needs

→

Make an Impact
Open source for a better future

Open Source • Open Minds • Endless Possibilities

1. Introduction to Open Source Software Development

* Introduction
* Benefits
* Challenges
* Open Source Products

Intended Learning Outcomes

* After the completion of this module, the student will be able to Explain and elaborate open-source software development.
  - Define Open Source Software (OSS)
  - Understand Open Source Software Development
  - Explain the benefits of OSS
  - Identify the challenges of OSS
  - Recognize popular open-source products

What is Open Source Software

* **Definition**

Open Source Software (OSS) is software whose source code is freely available for anyone to:

- View
- Modify
- Improve
- Distribute

* **Key Principle**

"Users have the freedom to use, study, modify, and share the software."

What is Source Code

* Source code is the human-readable instructions written by programmers.

* Examples of programming languages:
  - Python
  - Java
  - C++
  - JavaScript

* Without source code:
  - Users can only use the software.

* With source code:
  - Users can improve and customize it.

Introduction to Open Source Software Development

* A collaborative process where developers worldwide contribute to software projects.

* **Development Process**
  - Planning
  - Coding
  - Testing
  - Code Review
  - Documentation
  - Release
  - Community Feedback
  - Continuous Improvement

* Popular collaboration platforms:
  - GitHub
  - GitLab
  - Bitbucket

Characteristics of Open Source Software

- Free access to source code
- Community collaboration
- Transparency
- Continuous improvement
- Peer review
- Security through open inspection
- Flexible licensing

Benefits of Open Source Software

* **Technical Benefits**
  - High-quality software
  - Faster bug fixing
  - Better security
  - Frequent updates

* **Economic Benefits**
  - No licensing cost
  - Reduced software expenses
  - Lower maintenance costs

* **Educational Benefits**
  - Learning opportunity
  - Skill development
  - Community support

Challenges of Open Source Software

- Limited official support
- Compatibility issues
- Learning curve
- Documentation may vary
- Security risks if not maintained
- Sustainability of projects
- Dependency management

Popular Open Source Products

Product	Purpose
Linux	Operating System
Firefox	Web Browser
LibreOffice	Office Suite
VLC	Media Player
GIMP	Image Editing
Blender	3D Graphics
WordPress	Website Development
PostgreSQL	Database
MySQL	Database
Apache HTTP Server	Web Server

Advantages vs Challenges

Advantages	Challenges
Free to use	Limited support
Customizable	Learning curve
Secure	Compatibility issues
Community driven	Project sustainability
Transparent	Maintenance responsibility

Real-World Applications

* Open source software is widely used in:
  - Education
  - Government
  - Banking
  - Healthcare
  - Cloud Computing
  - Artificial Intelligence
  - Mobile Devices (Android)
  - Scientific Research
  - Web Development

* Examples:
  - Linux servers
  - Android OS
  - Apache Web Server
  - Moodle LMS

Summary

* **Key Points**

✓ Open Source Software provides access to source code.
✓ It encourages collaboration and innovation.

✓ Benefits include:
- Cost savings
- Flexibility
- Security
- Learning opportunities

✓ Challenges include:
- Support
- Compatibility
- Maintenance

✓ Many popular software products today are open source.

Questions

1. What is Open Source Software?
2. Name three benefits of OSS.
3. What are two challenges of OSS?
4. Name four popular open-source products.
5. Why do organizations use open-source software?

THANK YOU